

For the use of a Registered Medical Practitioner only

PRESCRIBING INFORMATION

ZETEZE-SM 10+10, 10+20, 10+40, 10+80 mg (Ezetimibe and Simvastatin Tablets)

COMPOSITION

ZETEZE-SM 10+10 mg Tablets

Each uncoated tablet contains

Ezetimibe USP10 mg

Simvastatin Ph. Eur.....10 mg

ZETEZE-SM 10+20 mg Tablets

Each uncoated tablet contains

Ezetimibe USP10 mg

Simvastatin Ph. Eur.....20 mg

ZETEZE-SM 10+40 mg Tablets

Each uncoated tablet contains

Ezetimibe USP10 mg

Simvastatin Ph. Eur.....40 mg

ZETEZE-SM 10+80 mg Tablets

Each uncoated tablet contains

Ezetimibe USP10 mg

Simvastatin Ph. Eur.....80 mg

Excipients: Lactose Monohydrate, Croscarmellose Sodium, Hypromellose, Cellulose Microcrystalline, Citric Acid Monohydrate, Propyl Gallate, Butylated Hydroxy Anisole, Magnesium Stearate, Purified Water and Isopropyl Alcohol.

PRODUCT DESCRIPTION

ZETEZE-SM 10+10 mg are white to off white, capsule shaped uncoated tablets, debossed with 'L' on one side and plain on other side.

ZETEZE-SM 10+20 mg are white to off white, capsule shaped uncoated tablets, debossed with 'I' on one side and plain on other side.

ZETEZE-SM 10+40 mg are white to off white, capsule shaped uncoated tablets, debossed with 'F' on one side and plain on other side.

ZETEZE-SM 10+80 mg are white to off white, capsule shaped uncoated tablets, debossed with 'T' on one side and plain on other side.

DESCRIPTION

ZETEZE-SM (ezetimibe/simvastatin) is a lipid lowering product that selectively inhibits the intestinal absorption of cholesterol and related plant sterols and inhibits the endogenous synthesis of cholesterol.

PHARMACODYNAMIC AND PHARMACOKINETIC PROPERTIES

- **Pharmacodynamics**

Mechanism of action:

Ezetimibe and Simvastatin

Plasma cholesterol is derived from intestinal absorption and endogenous synthesis. Ezetimibe and simvastatin are two lipid-lowering compounds with complementary mechanisms of action. Ezetimibe and simvastatin reduces elevated total cholesterol (total-C), LDL-C, apolipoprotein B (Apo B), triglycerides (TG), and non-high-density lipoprotein cholesterol (non-HDL-C), and increases high-density lipoprotein cholesterol (HDL-C) through dual inhibition of cholesterol absorption and synthesis.

Ezetimibe

Ezetimibe inhibits the intestinal absorption of cholesterol. Ezetimibe is orally active and has a mechanism of action that differs from other classes of cholesterol-reducing compounds (e.g., statins, bile acid sequestrants [resins], fibric acid derivatives, and plant stanols). The molecular target of ezetimibe is the sterol transporter, Niemann-Pick C1-Like 1 (NPC1L1), which is responsible for the intestinal uptake of cholesterol and phytosterols.

Ezetimibe localises at the brush border of the small intestine and inhibits the absorption of cholesterol, leading to a decrease in the delivery of intestinal cholesterol to the liver; statins reduce cholesterol synthesis in the liver and together these distinct mechanisms provide complementary cholesterol reduction.

Ezetimibe inhibited the absorption of cholesterol with no effect on the absorption of triglycerides, fatty acids, bile acids, progesterone, ethinyl estradiol, or fat soluble vitamins A and D.

Simvastatin

After oral ingestion, simvastatin, which is an inactive lactone, is hydrolysed in the liver to the corresponding active β -hydroxyacid form which has a potent activity in inhibiting HMG-CoA reductase (3 hydroxy - 3 methylglutaryl CoA reductase). This enzyme catalyses the conversion of HMG-CoA to mevalonate, an early and rate-limiting step in the biosynthesis of cholesterol.

Simvastatin has been shown to reduce both normal and elevated LDL-C concentrations. LDL is formed from very-low-density protein (VLDL) and is catabolised predominantly by the high affinity LDL receptor. The mechanism of the LDL-lowering effect of simvastatin may involve both reduction of VLDL-cholesterol (VLDL-C) concentration and induction of the LDL receptor, leading to reduced production and increased catabolism of LDL-C. Apolipoprotein B also falls substantially during treatment with simvastatin. In addition, simvastatin moderately increases HDL-C and reduces plasma TG. As a result of these changes, the ratios of total- to HDL-C and LDL- to HDL-C are reduced.

• Pharmacokinetics

Absorption

Ezetimibe

After oral administration, ezetimibe is rapidly absorbed and extensively conjugated to a pharmacologically active phenolic glucuronide (ezetimibe-glucuronide). Mean maximum plasma concentrations (C_{max}) occur within 1 to 2 hours for ezetimibe-glucuronide and 4 to 12 hours for ezetimibe. The absolute bioavailability of ezetimibe cannot be determined as the compound is virtually insoluble in aqueous media suitable for injection.

Concomitant food administration (high fat or non-fat meals) had no effect on the oral bioavailability of ezetimibe when administered as 10-mg tablets.

Simvastatin

The availability of the β -hydroxyacid to the systemic circulation following an oral dose of simvastatin was found to be less than 5% of the dose, consistent with extensive hepatic first-pass extraction. The major metabolites of simvastatin present in human plasma are the β -hydroxyacid and four additional active metabolites.

Relative to the fasting state, the plasma profiles of both active and total inhibitors were not affected when simvastatin was administered immediately before a test meal.

Distribution

Ezetimibe

Ezetimibe and ezetimibe-glucuronide are bound 99.7% and 88 to 92% to human plasma proteins, respectively.

Simvastatin

Both simvastatin and the β -hydroxyacid are bound to human plasma proteins (95%). No accumulation has been reported after single and multiple doses of simvastatin. The maximum plasma concentration of inhibitors was reported to occur 1.3 to 2.4 hours post-dose.

Biotransformation

Ezetimibe

Ezetimibe is metabolised primarily in the small intestine and liver via glucuronide conjugation (a phase II reaction) with subsequent biliary excretion. Minimal oxidative metabolism (a phase I reaction) has been reported with ezetimibe. Ezetimibe and ezetimibe-glucuronide are the major drug-derived compounds detected in plasma, constituting approximately 10 to 20% and 80 to 90% of the total drug in plasma, respectively. Both ezetimibe and ezetimibe-glucuronide are slowly eliminated from plasma with evidence of significant enterohepatic recycling. The half-life for ezetimibe and ezetimibe-glucuronide is approximately 22 hours.

Simvastatin

Simvastatin is an inactive lactone which is readily hydrolyzed *in vivo* to the corresponding β -hydroxyacid, a potent inhibitor of HMG-CoA reductase. Hydrolysis takes place mainly in the liver; the rate of hydrolysis in human plasma is very slow.

In man, simvastatin is well absorbed and undergoes extensive hepatic first-pass extraction. The extraction in the liver is dependent on the hepatic blood flow. The liver is its primary site of action, with subsequent excretion of drug equivalents in the bile. Consequently, availability of active drug to the systemic circulation is low.

Following an intravenous injection of the β -hydroxyacid metabolite, its half-life averaged 1.9 hours.

Elimination

Ezetimibe

Ezetimibe is excreted primarily in the faeces. After an oral dose, about 78% is excreted in the faeces, mainly as ezetimibe, and about 11% is excreted in the urine.

Simvastatin

Simvastatin acid is taken up actively into the hepatocytes by the transporter OATP1B1. Simvastatin is mainly excreted in the faeces via the bile as metabolites. About 10 to 15% is recovered in the urine, mainly in inactive forms. The half-life of the active β -hydroxyacid metabolite is 1.9 hours.

Special Populations

Paediatric Population

The absorption and metabolism of ezetimibe are similar between children and adolescents (10 to 18 years) and adults. Based on total ezetimibe, there are no pharmacokinetic differences between adolescents and adults. Pharmacokinetic data in the paediatric population <10 years of age are not available. Clinical experience in paediatric and adolescent patients (ages 9 to 17) includes patients with HoFH, HeFH, or sitosterolaemia (see **DOSE AND METHOD OF ADMINISTRATION**).

Elderly

Plasma concentrations for total ezetimibe are about 2-fold higher in the elderly (≥ 65 years) than in the young (18 to 45 years). LDL-C reduction and safety profile are comparable between elderly and younger subjects treated with ezetimibe (see **DOSE AND METHOD OF ADMINISTRATION**).

Hepatic impairment

After a single 10-mg dose of ezetimibe, the mean AUC for total ezetimibe was increased approximately 1.7-fold in patients with mild hepatic impairment (Child- Pugh score 5 or 6), compared to healthy subjects. On multiple-dosing (10 mg daily) in patients with moderate hepatic impairment (Child-Pugh score 7 to 9), the mean AUC for total ezetimibe was increased approximately 4-fold on Day 1 and Day 14 compared to healthy subjects. No dosage adjustment is necessary for patients with mild hepatic impairment. Due to the unknown effects of the increased exposure to ezetimibe in patients with moderate or severe (Child-Pugh score >9) hepatic impairment, ezetimibe is not recommended in these patients (see **DOSE AND METHOD OF ADMINISTRATION** and **WARNINGS AND PRECAUTIONS**).

Renal impairment

Ezetimibe

After a single 10-mg dose of ezetimibe in patients with severe renal disease, mean CrCl ≤ 30 ml/min), the mean AUC for total ezetimibe was increased approximately 1.5-fold, compared to healthy subjects (see **DOSE AND METHOD OF ADMINISTRATION**).

An additional patient (post-renal transplant and receiving multiple medications, including ciclosporin) had a 12-fold greater exposure to total ezetimibe.

Simvastatin

In patients with severe renal impairment (creatinine clearance <30 ml/min), the plasma concentrations of total inhibitors after a single dose of a related HMG-CoA reductase inhibitor were approximately two-fold higher than those in healthy subjects.

Gender

Plasma concentrations for total ezetimibe are slightly higher (approximately 20%) in women than in men. LDL-C reduction and safety profile are comparable between men and women treated with ezetimibe.

Race

There were no pharmacokinetic differences between Blacks and Caucasians.

INDICATIONS

Prevention of Cardiovascular Events

Ezetimibe and simvastatin is indicated to reduce the risk of cardiovascular events (cardiovascular death, nonfatal myocardial infarction, nonfatal stroke, hospitalization for unstable angina, or need for revascularization), in patients with coronary heart disease (CHD) and history of acute coronary syndrome (ACS), either previously treated with a statin or not.

Primary hypercholesterolaemia

Ezetimibe and simvastatin is indicated as adjunctive therapy to diet for the reduction of elevated total cholesterol (total-C), low-density lipoprotein cholesterol (LDL-C), apolipoprotein B (Apo B), triglycerides (TG), and non-high density lipoprotein cholesterol (non-HDL-C), and to increase high-density lipoprotein cholesterol (HDL-C) in patients with primary (heterozygous familial and non-familial) hypercholesterolaemia or mixed hyperlipidemia.

Homozygous Familial Hypercholesterolaemia (HoFH)

Ezetimibe and simvastatin is indicated for the reduction of elevated total-C and LDL-C levels in patients with HoFH. Patients may also receive adjunctive treatments (e.g., LDL apheresis).

DOSE AND METHOD OF ADMINISTRATION

The patient should be placed on a standard cholesterol-lowering diet before receiving ezetimibe and simvastatin and should continue on this diet during treatment with ezetimibe and simvastatin. The dosage should be individualized according to the baseline LDL-C level, the recommended goal of therapy, and the patient's response.

Instructions for Use: Ezetimibe and simvastatin should be taken as a single daily dose in the evening, with or without food.

Patients with Primary Hypercholesterolemia

In patients with primary hyperlipidemia or mixed hyperlipidemia, the dosage range of ezetimibe and simvastatin is 10/10 mg/day through 10/80 mg/day. The recommended usual starting dose is 10/20 mg/day. Initiation of therapy with 10/10 mg/day may be considered for patients requiring less aggressive LDL-C reductions.

Patients who require a larger reduction in LDL-C (greater than 55%) may be started as 10/40 mg/day. After initiation or titration of ezetimibe and simvastatin, lipid level may be analyzed after 2 or more weeks and dosage adjusted, if needed. The 10/80 mg dose of ezetimibe and simvastatin is only recommended in patients at high risk for cardiovascular complications who have not achieved their treatment goals on lower doses and when the benefits are expected to outweigh the potential risks (see **WARNINGS AND PRECAUTIONS**).

For patients who are currently on both simvastatin and ezetimibe, no dosage adjustment is required when switching to ezetimibe and simvastatin combination.

Dosage in Patients with Homozygous Familial Hypercholesterolaemia

The recommended dosage for patients with homozygous familial hypercholesterolaemia is ezetimibe and simvastatin 10/40 mg/day or 10/80 mg/day in the evening. The 10/80-mg dose is only recommended when the benefits are expected to outweigh the potential risks (see **CONTRAINDICATIONS** and **WARNINGS AND PRECAUTIONS**). Ezetimibe and simvastatin should be used as an adjunct to other lipid-lowering treatments (e.g., LDL apheresis) in these patients or if such treatments are unavailable.

In patients taking lomitapide concomitantly with ezetimibe and simvastatin, the dose of **ZETEZE-SM** should not exceed 10/40 mg/day.

Patients with Coronary Heart Disease

The 10/80 mg dose is only recommended in patients at high risk for cardiovascular complications who have not achieved treatment goals on lower doses and when the benefits are expected to outweigh the potential risks.

Use in the Elderly

No dosage adjustment is required for elderly patients (see **Pharmacokinetics**).

Use in Paediatric Patients

Treatment with ezetimibe and simvastatin is not recommended.

Use in Hepatic Impairment

No dosage adjustment is required in patients with mild hepatic insufficiency (Child-Pugh score 5 to 6). Treatment with ezetimibe and simvastatin is not recommended in patients with moderate (Child-Pugh score 7 to 9) or severe (Child-Pugh score >9) liver dysfunction (see **WARNINGS AND PRECAUTIONS** and **Pharmacokinetics**).

Use in Renal Impairment

No dosage adjustment is required for patients with moderate renal impairment. If treatment in patients with severe renal insufficiency (creatinine clearance ≤ 30 mL/min) is deemed necessary, dosages above 10/10 mg/day should be implemented cautiously (see **Pharmacokinetics**).

Co-administration with other medicines

Dosing of ezetimibe and simvastatin should occur either ≥ 2 hours before or ≥ 4 hours after administration of a bile acid sequestrant.

In patients taking amiodarone, verapamil or diltiazem, or products containing elbasvir or grazoprevir concomitantly with **ZETEZE-SM**, the dose of **ZETEZE-SM** should not exceed 10/20 mg/day (see **WARNINGS AND PRECAUTIONS** and **DRUG INTERACTIONS**).

In patients taking amlodipine or lipid-lowering dose of niacin (≥ 1 g/day) concomitantly with **ZETEZE-SM**, the dose of **ZETEZE-SM** should not exceed 10/40mg/day.

The safety and effectiveness of **ZETEZE-SM** administered with fibrates have not been reported. Therefore, the combination of **ZETEZE-SM** and fibrates should be avoided.

CONTRAINDICATIONS

- Hypersensitivity to the active substance(s) or to any of the excipients.
- Pregnancy and nursing.
- Active liver disease or unexplained persistent elevations in serum transaminases.
- Concomitant administration of potent CYP3A4 inhibitors (agents that increase AUC approximately 5 fold or greater) (e.g., itraconazole, ketoconazole, posaconazole, voriconazole, erythromycin, clarithromycin, telithromycin, HIV protease inhibitors (e.g. nelfinavir), boceprevir, telaprevir, nefazodone, and drugs containing cobicistat) (see **WARNINGS AND PRECAUTIONS** and **DRUG INTERACTIONS**).
- Concomitant administration of gemfibrozil, ciclosporin, or danazol (see **WARNINGS AND PRECAUTIONS** and **DRUG INTERACTIONS**).

WARNINGS AND PRECAUTIONS

Myopathy/Rhabdomyolysis

Simvastatin, like other inhibitors of HMG-CoA reductase, occasionally causes myopathy manifested as muscle pain, tenderness or weakness with creatine kinase (CK) above 10X the upper limit of normal (ULN). Myopathy sometimes takes the form of rhabdomyolysis with or without acute renal failure secondary to myoglobinuria, and rare fatalities have been reported. The risk of myopathy is increased by high levels of HMG-CoA reductase inhibitory activity in plasma.

Predisposing factors for myopathy include advanced age (≥ 65 years), female gender, uncontrolled hypothyroidism, renal impairment, personal or familial history of hereditary muscular disorders, previous history of muscular toxicity with a statin or fibrate and alcohol abuse.

In such situations, the risk of treatment should be considered in relation to possible benefit, and clinical monitoring is recommended. If a patient has previously experienced a muscle disorder on a fibrate or a statin, treatment with any statin containing product (such as ezetimibe and simvastatin) should only be initiated with caution. If CK levels are significantly elevated at baseline ($>5 \times \text{ULN}$), treatment should not be started.

As with other HMG-CoA reductase inhibitors, the risk of myopathy/rhabdomyolysis is dose related for simvastatin.

The risk of myopathy is greater in patients on ezetimibe and simvastatin 10/80 mg compared with other statin-based therapies with similar LDL-C-lowering efficacy. Therefore, the 10/80-mg dose of ezetimibe and simvastatin should only be used in patients with severe hypercholesterolaemia and at high risk for cardiovascular complications who have not achieved their treatment goals on lower doses and when the benefits are expected to outweigh the potential risks. In patients taking ezetimibe and simvastatin 10/80 mg for whom an interacting agent is needed, a lower dose of ezetimibe and simvastatin or an alternative statin-based regimen with less potential for drug-drug interactions should be used.

All patients starting therapy with ezetimibe and simvastatin, or whose dose of ezetimibe and simvastatin is being increased, should be advised of the risk of myopathy and told to report promptly and unexplained muscle pain, tenderness or weakness. Ezetimibe and simvastatin therapy should be discontinued immediately if myopathy is diagnosed or suspected. The presence of these symptoms, and a CK level > 10 times the upper limit of normal indicates myopathy. In most cases, when patients were promptly discontinued from simvastatin treatment, muscle symptoms and CK increases resolved. Periodic CK determination may be considered in patients starting therapy with ezetimibe and simvastatin or whose dose is being increased. Periodic CK determinations are recommended for patients titrating to the 10/80 mg dose. There is no assurance that such monitoring will prevent myopathy.

Many of the patients who have developed rhabdomyolysis on therapy with simvastatin have had complicated medical histories, including renal impairment usually as a consequence of long-standing diabetes mellitus. Such patients taking ezetimibe and simvastatin merit closer monitoring. Therapy with ezetimibe and simvastatin should be temporarily stopped a few days prior to elective major surgery and when any major medical or surgical condition supervenes.

Caution should be used when prescribing ezetimibe and simvastatin to Asian patients and the lowest dose necessary should be employed.

Creatine Kinase measurement

Creatine Kinase (CK) should not be measured following strenuous exercise or in the presence of any plausible alternative cause of CK increase as this makes value interpretation difficult. If CK levels are significantly elevated at baseline ($>5 \times \text{ULN}$), levels should be re-measured within 5 to 7 days later to confirm the results.

If muscle pain, weakness or cramps occur whilst a patient is receiving treatment with ezetimibe and simvastatin, their CK levels should be measured and if these levels are found to be significantly elevated ($>5 \times \text{ULN}$), treatment should be stopped. If muscular symptoms are severe and cause daily discomfort, even if CK levels are $<5 \times \text{ULN}$, treatment discontinuation may be considered. If myopathy is suspected for any other reason, treatment should be discontinued.

There have been very rare reports of an immune-mediated necrotizing myopathy (IMNM) during or after treatment with some statins. IMNM is clinically characterized by:

- persistent proximal muscle weakness and elevated serum creatine kinase, which persist despite discontinuation of statin treatment;
- muscle biopsy showing necrotizing myopathy without significant inflammation;
- improvement with immunosuppressive agents.

If symptoms resolve and CK levels return to normal, then reintroduction of ezetimibe and simvastatin or introduction of another statin containing product may be considered at the lowest dose and with close monitoring.

Reduced function of transport proteins

Reduced function of hepatic OATP transport proteins can increase the systemic exposure of simvastatin acid and increase the risk of myopathy and rhabdomyolysis. Reduced function can occur as the result of inhibition by interacting medicines (eg ciclosporin) or in patients who are carriers of the SLCO1B1 c.521T>C genotype.

Patients carrying the SLCO1B1 gene allele (c.521T>C) coding for a less active OATP1B1 protein have an increased systemic exposure of simvastatin acid and increased risk of myopathy. The risk of high dose (80 mg) simvastatin related myopathy is about 1 % in general, without genetic testing.

Homozygote C allele carriers (also called CC) treated with 80 mg have a 15% risk of myopathy within one year, while the risk in heterozygote C allele carriers (CT) is 1.5%. The corresponding risk is 0.3% in patients having the most common genotype (TT). Where available, genotyping for the presence of the C allele should be considered as part of the benefit-risk assessment prior to prescribing 80 mg simvastatin for individual patients and high doses avoided in those found to carry the CC genotype. However, absence of this gene upon genotyping does not exclude that myopathy can still occur.

Liver Enzymes

In patients receiving ezetimibe and simvastatin concomitantly, consecutive transaminase elevations (≥ 3 X ULN) have been reported (see **SIDE EFFECTS/UNDESIRABLE EFFECTS**).

It is recommended that liver function tests be performed before treatment with ezetimibe and simvastatin begins and thereafter when clinically indicated. Patients titrated to the 10/80-mg dose should receive an additional test prior to titration, 3 months after titration to the 10/80-mg dose, and periodically thereafter (e.g., semiannually) for the first year of treatment. Special attention should be paid to patients who develop elevated serum transaminase levels, and in these patients, measurements should be repeated promptly and then performed more frequently. If the transaminase levels show evidence of progression, particularly if they rise to 3 X ULN and are persistent, the drug should be discontinued. Note that ALT may emanate from muscle, therefore ALT rising with CK may indicate myopathy (see above *Myopathy/Rhabdomyolysis*).

There have been rare post-marketing reports of fatal and non-fatal hepatic failure in patients taking statins, including simvastatin. If serious liver injury with clinical symptoms and/or hyperbilirubinaemia or jaundice occurs during treatment with ezetimibe and simvastatin promptly interrupt therapy. If an alternate etiology is not found, do not restart ezetimibe and simvastatin.

Ezetimibe and simvastatin should be used with caution in patients who consume substantial quantities of alcohol and/or have a past history of liver disease. Active liver diseases or unexplained persistent transaminase elevations are contraindications to the use of ezetimibe and simvastatin.

Hepatic impairment

Due to the unknown effects of the increased exposure to ezetimibe in patients with moderate or severe hepatic impairment, ezetimibe and simvastatin is not recommended in these patients (see **Pharmacokinetics**).

Diabetes mellitus

Some evidence suggests that statins as a class raise blood glucose and in some patients, at high risk of future diabetes, may produce a level of hyperglycaemia where formal diabetes care is appropriate. This risk, however, is outweighed by the reduction in vascular risk with statins and therefore should not be a reason for stopping statin treatment. Patients at risk (fasting glucose 5.6 to 6.9 mmol/L, BMI

> 30 kg/m², raised triglycerides, hypertension) should be monitored both clinically and biochemically according to national guidelines.

Interstitial lung disease

Cases of interstitial lung disease have been reported with some statins, including simvastatin, especially with long term therapy. Presenting features can include dyspnoea, non-productive cough and deterioration in general health (fatigue, weight loss and fever). If it is suspected a patient has developed interstitial lung disease, ezetimibe and simvastatin therapy should be discontinued.

Effects on ability to drive and use machines

No studies on the effects on the ability to drive and use machines have been reported. However, when driving vehicles or operating machines, it should be taken into account that dizziness has been reported.

Lactose

ZETEZE-SM contains lactose. Patients with rare hereditary problems of galactose intolerance, the Lapp lactase deficiency or glucose-galactose malabsorption should not take this medicine.

DRUG INTERACTIONS

Contraindicated drugs

Simvastatin is a substrate of cytochrome P450 3A4. Potent inhibitors of cytochrome P450 3A4 increase the risk of myopathy and rhabdomyolysis by increasing the concentration of HMGCoA reductase inhibitory activity in plasma during simvastatin therapy.

Potent inhibitors of CYP 3A4: Concomitant use with medicines labeled as having a potent inhibitory effect on CYP3A4 at therapeutic doses (e.g. itraconazole, ketoconazole, posaconazole, voriconazole, erythromycin, clarithromycin, telithromycin, HIV protease inhibitors, boceprevir, telaprevir, nefazodone or medicinal products containing cobicistat) is contraindicated. If short term treatment with potent CYP3A4 inhibitors is unavoidable, therapy with ezetimibe and simvastatin should be suspended during the course of treatment (see **CONTRAINDICATIONS**).

Gemfibrozil, ciclosporin or danazol: Concomitant use of these drugs with ezetimibe and simvastatin is contraindicated (see **CONTRAINDICATIONS**).

- Gemfibrozil: Gemfibrozil increases the AUC of simvastatin acid by 1.9 fold, possibly due to inhibition of the glucuronidation pathway and/or OATP1B1.

- Ciclosporin: The risk of myopathy/rhabdomyolysis is increased by concomitant administration of ciclosporin with ezetimibe and simvastatin,. Although the mechanism is not fully understood, ciclosporin has been shown to increase the AUC of HMGCoA reductase inhibitors. The increase in AUC for simvastatin acid is presumably due, in part, to inhibition of CYP3A4 and/or OATP1B1.
- Danazol: The risk of myopathy and rhabdomyolysis is increased by concomitant administration of danazol with ezetimibe and simvastatin;.

Other drug interactions

Other fibrates: The dose of simvastatin should not exceed 10 mg daily in patients receiving concomitant medication with fibrates other than gemfibrozil or fenofibrate. When simvastatin and fenofibrate are given concomitantly, there is no evidence that the risk of myopathy exceeds the sum of the individual risks of each agent. Caution should be used when prescribing fenofibrate with simvastatin, as either agent can cause myopathy when given alone. Addition of fibrates to simvastatin typically provides little additional reduction in LDL-C, but further reductions of TG and further increases in HDL-C may be obtained. Combinations of fibrates with simvastatin have been reported to be used without myopathy in small short-term clinical studies with careful monitoring.

Concurrent use of fibrates may cause severe myositis and myoglobinuria.

Niacin: In healthy adults, concomitant ezetimibe and simvastatin (10/20 mg daily for 7 days) caused a small increase in the mean AUCs of niacin (22%) and nicotinuric acid (19%) administered as niacin extended-release tablets (1000 mg for 2 days and 2000 mg for 5 days following a low-fat breakfast). Concomitant niacin slightly increased the mean AUCs of ezetimibe (9%), total ezetimibe (26%), simvastatin (20%) and simvastatin acid (35%). These increases are not considered clinically significant.

Cholestyramine: Concomitant cholestyramine administration decreased the mean area under the curve (AUC) of total ezetimibe (ezetimibe + ezetimibe glucuronide) approximately 55%. The incremental LDL-C reduction due to adding ezetimibe and simvastatin to cholestyramine may be lessened by this interaction.

Colchicine: Cases of myopathy and rhabdomyolysis have been reported with the concomitant administration of colchicine and ezetimibe and simvastatin in patients with renal impairment. Close clinical monitoring of such patients taking this combination is advised.

Antacids: Concomitant antacid administration decreased the rate of absorption of ezetimibe but had no effect on the bioavailability of ezetimibe. This decreased rate of absorption is not considered clinically significant.

Fusidic acid: The risk of myopathy including rhabdomyolysis may be increased by the concomitant administration of systemic fusidic acid with statins. Co-administration of this combination may cause increased plasma concentrations of both agents. The mechanism of this interaction (whether it is pharmacodynamics or pharmacokinetic, or both) is yet unknown. There have been reports of rhabdomyolysis (including some fatalities) in patients receiving this combination. If treatment with fusidic acid is necessary, ezetimibe and simvastatin treatment should be discontinued throughout the duration of the fusidic acid treatment. In exceptional circumstances, where prolonged systemic fusidic acid is needed, e.g. for the treatment of severe infections, the need for co-administration of ezetimibe and simvastatin and fusidic acid should only be considered on a case-by-case basis under close medical supervision.

Amiodarone: Myopathy was reported in patients receiving simvastatin 80 mg and amiodarone. The dose of simvastatin should not exceed 20 mg daily in patients receiving concomitant medication with amiodarone.

Calcium channel blockers:

- *Verapamil or diltiazem:* It was reported that patients on diltiazem treated concomitantly with simvastatin 80 mg had an increased risk of myopathy. The dose of simvastatin should not exceed 20 mg daily in patients receiving concomitant medication with verapamil or diltiazem.
- *Amlodipine:* It was reported that patients on amlodipine treated concomitantly with simvastatin 80 mg had a slightly increased risk of myopathy. The dose of simvastatin should not exceed 40 mg daily in patients receiving concomitant medication with amlodipine.
- *Niacin ($\geq 1\text{g/day}$):* The dose of simvastatin should not exceed 40mg daily in patients receiving concomitant medication with niacin (nicotinic acid) $\geq 1\text{g/day}$. Cases of myopathy/rhabdomyolysis have been reported with simvastatin co-administered with lipid-modifying doses ($\geq 1\text{ g/day}$) of niacin.

Acipimox is structurally related to niacin. Although acipimox was not studied, the risk for muscle related toxic effects may be similar to niacin.

Lomitapide: The risk of myopathy and rhabdomyolysis may be increased by concomitant administration of lomitapide with simvastatin.

Moderate Inhibitors of CYP3A4: Patients taking other medicines labelled as having a moderate inhibitory effect on CYP3A4 concomitantly with ezetimibe and simvastatin, particularly higher ezetimibe and simvastatin doses, may have an increased risk of myopathy. When coadministering

ezetimibe and simvastatin with a moderate inhibitor of CYP3A4, a dose adjustment of ezetimibe and simvastatin may be necessary.

Inhibitors of the Transport Protein OATP1B1: Simvastatin acid is a substrate of the transport protein OATP1B1. Concomitant administration of medicinal products that are inhibitors of the transport protein OATP1B1 may lead to increased plasma concentrations of simvastatin acid and an increased risk of myopathy.

Other interactions: Grapefruit juice contains one or more components that inhibit CYP3A4 and can increase the plasma levels of drugs metabolized by CYP3A4. The effect of typical consumption (one 250-ml glass daily) is minimal (13% increase in active plasma HMG-CoA reductase inhibitory activity as measured by the area under the concentration-time curve) and of no clinical relevance. However, because larger quantities significantly increase the plasma levels of HMG-CoA reductase inhibitory activity grapefruit juice should be avoided during ezetimibe and simvastatin therapy.

Rifampicin: Because rifampicin is a potent CYP3A4 inducer, patients undertaking long-term rifampicin therapy (e.g. treatment of tuberculosis) may experience loss of efficacy of simvastatin.

Fluconazole: Rare cases of rhabdomyolysis associated with concomitant administration of simvastatin and fluconazole have been reported.

Anticoagulants: Simvastatin 20-40 mg/day modestly potentiated the effect of coumarin anticoagulants: the prothrombin time, reported as International Normalised Ratio (INR), increased from a baseline of 1.7 to 1.8 and from 2.6 to 3.4 in the normal subjects and hypercholesterolaemic patient, respectively. Very rare cases of elevated INR have been reported. In patients taking coumarin anticoagulants, prothrombin time should be determined before starting ezetimibe and simvastatin and frequently enough during early therapy to ensure that no significant alteration of prothrombin time occurs. Once a stable prothrombin time has been documented, prothrombin times can be monitored at the intervals usually recommended for patients on coumarin anticoagulants. If the dose of ezetimibe and simvastatin is changed or discontinued, the same procedure should be repeated. Simvastatin therapy has not been associated with bleeding or with changes in prothrombin time in patients not taking anticoagulants.

Concomitant administration of ezetimibe (10 mg once daily) had no significant effect on bioavailability of warfarin and prothrombin time in healthy adult males. However, there have been post-marketing reports of increased International Normalised Ratio (INR) in patients who had ezetimibe added to warfarin or fluindione. If ezetimibe and simvastatin is added to warfarin, another coumarin anticoagulant, or fluindione, INR should be appropriately monitored.

SIDE EFFECTS/UNDESIRABLE EFFECTS

The adverse events reported with simvastatin and ezetimibe are presented below.

Table: Adverse reactions reported with ezetimibe and simvastatin

System organ class	Adverse reactions	Frequency
Investigations	ALT and/or AST increased; blood CK increased	Common
	blood bilirubin increased; blood uric acid increased; gamma-glutamyltransferase increased; international normalised ratio increased; protein urine present; weight decreased	Uncommon
Nervous system disorders	dizziness; headache	Uncommon
Gastrointestinal disorders	abdominal pain; abdominal discomfort; abdominal pain upper; dyspepsia; flatulence; nausea; vomiting	Uncommon
Skin and subcutaneous tissue disorders	pruritus; rash	Uncommon
Musculoskeletal and connective tissue disorders	arthralgia; muscle spasms; muscular weakness; musculoskeletal discomfort; neck pain; pain in extremity	Uncommon
General disorders and administration site condition	asthenia; fatigue; malaise; oedema; peripheral	Uncommon
Psychiatric disorders	sleep disorder	Uncommon

Table: Adverse reactions reported with ezetimibe and simvastatin and at a greater incidence than statins

System organ class	Adverse reactions	Frequency
Investigations	ALT and/or AST increased	Common
	blood bilirubin increased; blood CK increased; gamma-glutamyltransferase increased	Uncommon
Nervous system disorders	headache; paraesthesia	Uncommon
Gastrointestinal disorders	abdominal distension; diarrhoea; dry mouth; dyspepsia; flatulence; gastrooesophageal reflux disease; vomiting	Uncommon
Skin and subcutaneous tissue disorders	pruritus; rash; urticaria	Uncommon
Musculoskeletal and connective tissue disorders	myalgia	Common
	arthralgia; back pain; muscle spasms; muscular weakness; musculoskeletal pain; pain in extremity	Uncommon
General disorders and administration site condition	asthenia; chest pain; fatigue; oedema	Uncommon

Table: Adverse reactions reported with ezetimibe and simvastatin and at a greater incidence than statins

System organ class	Adverse reactions	Frequency
site condition	peripheral	
Psychiatric disorders	insomnia	Uncommon

The additional adverse reactions have been reported with ezetimibe and simvastatin or post-marketing use with one of the individual components. The adverse reactions reported for ezetimibe and simvastatin are consistent with those previously reported with ezetimibe and/or simvastatin.

Investigations: liver function test abnormal.

Blood and lymphatic system disorders: thrombocytopaenia; anaemia

Nervous system disorders: peripheral neuropathy; memory impairment

Respiratory, thoracic and mediastinal disorders: cough; dyspnoea, interstitial lung disease (see **WARNINGS AND PRECAUTIONS**).

Gastro-intestinal disorders: constipation; pancreatitis; gastritis

Skin and subcutaneous tissue disorders: alopecia; erythema multiforme; hypersensitivity reactions, including rash, urticaria, anaphylaxis, angio-oedema.

Musculoskeletal and, connective tissue disorders: muscle cramps; myopathy (including myositis); rhabdomyolysis with or without acute renal failure; tendinopathy, sometimes complicated by rupture; immune-mediated necrotizing myopathy (IMNM) (frequency not known).

Metabolism and nutrition disorders: decreased appetite

Vascular disorders: hot flush; hypertension

General disorders and administration site conditions: pain

Hepato-biliary disorders: hepatitis/jaundice; fatal and non-fatal hepatic failure; cholelithiasis; cholecystitis

Reproductive system and breast disorders: erectile dysfunction

Psychiatric disorders: depression, insomnia

An apparent hypersensitivity syndrome has been reported rarely which has included some of the following features: angio-oedema, lupus-like syndrome, polymyalgia rheumatica, dermatomyositis, vasculitis, thrombocytopaenia, eosinophilia, red blood cell sedimentation rate increased, arthritis and arthralgia, urticaria, photosensitivity reaction, pyrexia, flushing, dyspnoea and malaise.

Increases in HbA1c and fasting serum glucose levels have been reported with statins. The risk of hyperglycemia, however, is outweighed by the reduction in vascular risk with statins.

There have been rare post-marketing reports of cognitive impairment (e.g., memory loss, forgetfulness, amnesia, memory impairment, confusion) associated with statin use. These cognitive issues have been reported for all statins. The reports are generally nonserious, and reversible upon statin discontinuation, with variable times to symptom onset (1 day to years) and symptom resolution (median of 3 weeks).

Laboratory Values:

The incidence of clinically important elevations in serum transaminases (ALT and/or AST ≥ 3 X ULN, consecutive) was 1.7% for patients treated with ezetimibe and simvastatin. These elevations were generally asymptomatic, not associated with cholestasis, and returned to baseline after discontinuation of therapy or with continued treatment.

Clinically important elevations of CK (≥ 10 X ULN) were seen in 0.2% of the patients treated with ezetimibe and simvastatin.

The following additional adverse events have been reported with some statins:

- Sleep disturbances, including nightmares
- Sexual dysfunction
- Diabetes mellitus: Frequency will depend on the presence or absence of risk factors (fasting blood glucose ≥ 5.6 mmol/L, BMI > 30 kg/m², raised triglycerides, history of hypertension).

USE IN SPECIAL POPULATIONS

• Pregnancy

Atherosclerosis is a chronic process, and ordinarily discontinuation of lipid-lowering drugs during pregnancy should have little impact on the long-term risk associated with primary hypercholesterolemia.

Ezetimibe and Simvastatin

Ezetimibe and Simvastatin is contraindicated during pregnancy and lactation.

The safety of simvastatin in pregnant women has not been established. Rare reports of congenital anomalies following intrauterine exposure to HMG-CoA reductase inhibitors have been reported. However, in an analysis of approximately 200 prospectively followed pregnancies exposed during the first trimester to simvastatin or another closely related HMG-CoA reductase inhibitor, the incidence of congenital anomalies was comparable to that seen in the general population. This number of pregnancies was statistically sufficient to exclude a 2.5-fold or greater increase in congenital anomalies over the background incidence.

Although there is no evidence that the incidence of congenital anomalies in offspring of patients taking simvastatin or another closely related HMG-CoA reductase inhibitor differs from that reported in the general population, maternal treatment with simvastatin may reduce the foetal levels of mevalonate which is a precursor of cholesterol biosynthesis. For this reason, ezetimibe and simvastatin must not be used in women who are pregnant, trying to become pregnant or suspect they are pregnant. Treatment with ezetimibe and simvastatin must be suspended for the duration of pregnancy or until it has been determined that the woman is not pregnant (see **CONTRAINDICATIONS**).

Ezetimibe

No reported clinical data are available on the use of ezetimibe during pregnancy. When ezetimibe was given with simvastatin, no teratogenic effects were reported in embryo-fetal development in pregnant rats. In pregnant rabbits, a low incidence of skeletal malformations was reported.

• **Lactation**

Ezetimibe and simvastatin is contraindicated during lactation. Ezetimibe was reported to be excreted in rats milk.

OVERDOSE

Ezetimibe and Simvastatin

In the event of an overdose, symptomatic and supportive measures should be employed.

Ezetimibe

Ezetimibe was well tolerated at dose of 50 mg/day to healthy individuals for up to 14 days, or 40 mg/day to patients with primary hypercholesterolaemia for up to 56 days.

A few cases of overdosage have been reported; most have not been associated with adverse experiences. Reported adverse experiences have not been serious.

A few cases of overdosage have been reported; the maximum dose taken was 3.6 g. All patients recovered without sequelae.

STORAGE

Store below 30°C.

SUPPLY

Cold form blister pack of 3 ×10's/box

Manufactured In India by:**Sun Pharmaceutical Industries Ltd.**

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